

The Language of Medicine

13th edition

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Chapter 21

Pharmacology

Chapter Goals (Slide 1 of 2)

- Describe the various subspecialty areas of pharmacology.
- Identify the various routes and schedule of drug administration.
- Differentiate among the various classes of drugs and name their primary actions and side effects.

Chapter Goals (Slide 2 of 2)

- Define medical terms using combining forms, prefixes, and suffixes that relate to pharmacology.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 21

Lesson 21.1

Introduction

- Drugs: medicines/substances used to prevent or treat a condition or disease
- Obtained from various sources:
 - Plants: for example, digitalis
 - Animals: for example, hormones
 - Synthesized in lab: for example, methotrexate

Pharmacology Professionals

- Pharmacist (PharmD)
- Pharmacologist (MD or PhD)
- Subspecialties include:
 - Medicinal chemistry (synthesize new drugs)
 - Molecular pharmacology (interaction of drugs and subcellular structures)
 - Chemotherapy (drugs that destroy microorganisms, parasites, malignant cells)
 - Toxicology (harmful effects of drugs and antidotes)

Biological Effects of Drugs

- **Pharmacodynamics:** study of a drug's effects in the body
- **Pharmacokinetics:** study of a drug's absorption, distribution into body compartments, metabolism, and excretion over a period

QUICK QUIZ (Slide 1 of 3)

1. Which term means below the skin?

- A. Intravenous
- B. Subcutaneous
- C. Intracavity
- D. Transdermal

Drug Names, Standards, and References

- **Chemical name:** exact chemical makeup
- **Generic name:** legal and scientific name
- **Brand name:** trademark of drug manufacturer

Drug Standards and References

- Food and Drug Administration (FDA)
- United States Pharmacopeia (USP)
- Hospital Formulary
- *Physicians' Desk Reference*[®] (*PDR*[®])

Administration of Drugs

- Oral
- Sublingual
- Rectal
- Parenteral:
 - Intracavitary injection
 - Intradermal injection
 - Intramuscular injection (IM)
 - Intrathecal injection
 - Intravenous injection
 - Subcutaneous injection (SC)
- Inhalation
- Topical

Terminology of Drug Action

- Receptor: target of drug
- Action and interaction of drugs in the body:
 - Combination of two drugs
 - Additive action
 - Antagonistic
 - Synergistic
- Other actions of drugs
 - Tolerance
 - Addiction

QUICK QUIZ (Slide 2 of 3)

2. Means that the combination of two drugs is greater than the total effects of each drug by itself:

- A. synergism (potentiation)
- B. tolerance
- C. additive action
- D. idiosyncrasy

Drug Toxicity

- Unwanted and potentially dangerous effects of drugs
- Idiosyncrasy (unexpected and uncommon; e.g., anaphylaxis)
- Iatrogenic (produced by treatment)
- Side effects (routine, often tolerable, reversible when drug is discontinued)
- Contraindications (dangerous and ill-advised use)
- Resistance (lack of beneficial response)

Chapter 21

Lesson 21.2

Classes of Drugs

- Analgesics
- Anesthetics
- Antibiotics and antivirals
- Anticoagulants
- Anticonvulsants
- Antidepressants
- Antidiabetics
- Antihistamines
- Antiosteoporosis
- Cardiovascular
- Endocrine
- Gastrointestinal
- Respiratory
- Sedative-hypnotics
- Stimulants
- Tranquilizers

Cardiovascular Drugs

- Angiotensin-converting enzyme (ACE) inhibitors
- Angiotensin II receptor antagonists
- Antiarrhythmics
- Beta-blockers
- Calcium antagonists or calcium channel blockers
- Cholesterol-lowering drugs
- Diuretics

Endocrine Drugs

- Androgens
- Antiandrogens
- Estrogens
- Antiestrogen
- Selective estrogen receptor modulator (SERM)
- Progestins
- Thyroid hormone
- Glucocorticoids

Gastrointestinal Drugs

- Antacids
- Antiulcer (e.g., histamine receptor blockers)
- Antidiarrheal
- Cathartics (laxatives or purgatives)
- Antinauseants (antiemetics)

Vocabulary (Slide 1 of 11)

Term	Meaning/Definition
addiction	Physical and psychological dependence on and craving for a drug
additive action	Drug action in which the combination of two similar drugs is equal to the sum of the effects of each
aerosol	Particles of drug suspended in air
anaphylaxis	Exaggerated hypersensitivity reaction to a previously encountered drug or foreign protein

Vocabulary (Slide 2 of 11)

Term	Meaning/Definition
antagonistic action	Combination of two drugs gives less than an additive effect
antidote	Agent given to counteract an unwanted effect of a drug
brand name	Commercial name for a drug; trademark or trade name
chemical name	Chemical formula for a drug

Vocabulary (Slide 3 of 11)

Term	Meaning/Definition
contraindications	Factors that prevent the use of a drug or treatment
controlled substances	Drugs that produce tolerance and dependence and have potential for abuse or addiction
dependence	Prolonged use of a drug that may lead to a physiologic need for its actions in the body
dose	Amount of drug administered

Vocabulary (Slide 4 of 11)

Term	Meaning/Definition
Food and Drug Administration (FDA)	U.S. government agency with legal responsibility to enforce proper drug manufacture and clinical use
generic name	Legal noncommercial name for a drug
iatrogenic	Condition caused by treatment given by physicians or medical personnel
idiosyncratic reaction	Unexpected effect produced in a particularly sensitive patient, but not seen in most people

Vocabulary (Slide 5 of 11)

Term	Meaning/Definition
inhalation	Administration of drugs in gaseous or vapor form through the nose or mouth
medicinal chemistry	Study of new drug synthesis
molecular pharmacology	Study of interaction of drugs and their target molecules
oral administration	Drugs are given by mouth

Vocabulary (Slide 6 of 11)

Term	Meaning/Definition
parenteral administration	Drugs are given by injection into the skin, muscles, or veins
pharmacist	Specialist in preparing and dispensing drugs
pharmacy	Location for preparing and dispensing drugs; also the study of preparing and dispensing drugs
pharmacodynamics	Study of drug effects within the body

Vocabulary (Slide 7 of 11)

Term	Meaning/Definition
pharmacokinetics	Study of drug absorption, distribution, metabolism, and excretion over a period of time
pharmacologist	Specialist in the study of properties, uses, and side effects of drugs
pharmacology	Study of the preparation, properties, uses, and side effects of drug
<i>Physicians' Desk Reference (PDR)</i>	Reference book that lists drug products

Vocabulary (Slide 8 of 11)

Term	Meaning/Definition
receptor	Target substance with which a drug interacts in the body
rectal administration	Drugs are inserted through the anus into the rectum
resistance	Lack of beneficial response
response	Desired and beneficial effect of a drug

Vocabulary (Slide 9 of 11)

Term	Meaning/Definition
schedule	Exact timing and frequency of drug administration
side effect	Adverse reaction, usually minor, that routinely results from use of a drug
sublingual administration	Drugs are given by placement under the tongue
synergism	Combination of two drugs causes an effect that is greater than the sum of the individual effects of each drug alone

Vocabulary (Slide 10 of 11)

Term	Meaning/Definition
syringe	Instrument for introducing or withdrawing fluids from the body
tolerance	Larger and larger drug doses must be given to achieve the desired effect
topical application	Drugs are applied locally on the skin or mucous membranes of the body
toxicity	Harmful effects of a drug

Vocabulary (Slide 11 of 11)

Term	Meaning/Definition
toxicology	Study of harmful chemicals and their effects on the body
transport	Movement of a drug across a cell membrane into body cells
United States Pharmacopeia (USP)	Authoritative list of drugs, formulas, and preparations that sets a standard for drug manufacturing and dispensing
vitamin	Substance found in foods and essential in small quantities for growth and good health

Combining Forms and Terminology (Slide 1 of 3)

Combining Form	Meaning
aer/o	air
algēs/o	sensitivity to pain
bronch/o	bronchial tube
chem/o	drug
cras/o	mixture
cutane/o	skin
derm/o	skin
erg/o	work

Combining Forms and Terminology (Slide 2 of 3)

Combining Form	Meaning
esthes/o	feeling, sensation
hist/o	tissue
hypn/o	sleep
iatr/o	treatment
lingu/o	tongue
myc/o	mold, fungus
narc/o	stupor
or/o	mouth

Combining Forms and Terminology (Slide 3 of 3)

Combining Form	Meaning
pharmac/o	drug
prurit/o	itching
pyret/o	fever
thec/o	sheath (of brain and spinal cord)
tox/o	poison
toxic/o	poison
vas/o	vessel
ven/o	vein
vit/o	life

Prefixes and Terminology

Prefix	Meaning
ana-	upward, excessive
anti-	against
contra-	against, opposite
par-	other than, apart
syn-	together, with

Abbreviations (Slide 1 of 5)

Abbreviation	Meaning
a.c., ac	Before meals
ACE	Angiotensin-converting enzyme
ad lib	Freely, as desired
APAP	Acetaminophen
ARB	Angiotensin II receptor blocker
b.i.d., bid	Two times a day
c-	With
Caps	Capsules
Cc	Cubic centimeter
FDA	U.S. Food and Drug Administration

Abbreviations (Slide 2 of 5)

Abbreviation	Meaning
gm, g	Gram
gtt	Drops
h	Hour
h.s., hs	At bedtime
H ₂ blocker	Histamine h ₂ receptor antagonist
HRT	Hormone replacement therapy
IM	Intramuscular
INH	Isoniazid-antituberculosis agent
IV	Intravenous
MAOI	Monoamine oxidase inhibitor

Abbreviations (Slide 3 of 5)

Abbreviation	Meaning
mg	Milligram
ml, mL	Milliliter
NPO	Nothing by mouth
NSAID	Nonsteroidal anti-inflammatory drug
p-	After
p.c., pc	After meals
PCA	Patient-controlled analgesia
PDR	Physicians' Desk Reference
PO, p.o. po	By mouth
p.r.n., prn	As needed, as necessary

Abbreviations (Slide 4 of 5)

Abbreviation	Meaning
Pt	Patient
q	Every
q.h., qh	Every hour
q2h	Every 2 hours
q.i.d., qid	Four times a day
q.s., qs	Sufficient quantity
qAM	Every morning
qPM	Every evening
Rx	Prescription
s-	Without

Abbreviations (Slide 5 of 5)

Abbreviation	Meaning
SERM	Selective estrogen receptor modulator
Sig.	Directions – how to take medication
SL	Sublingual
s.o.s.	If it is necessary (<i>si opus sit</i>)
SSRI	Selective serotonin reuptake inhibitor
SQ	Subcutaneous
tab	Tablet
TCA	Tricyclic antidepressant
t.i.d.,tid	Three times daily (<i>ter in die</i>)

QUICK QUIZ (Slide 3 of 3)

3. Which term means prevents nausea and vomiting?

- A. Antibiotic
- B. Anticoagulant
- C. Antiplatelet
- D. Antiemetic